

Targeted interventions and their effect on recovery in children, adolescents and adults who have sustained a sport-related concussion: a systematic review

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ABSTRACT

Objectives We evaluated interventions to facilitate recovery in children, adolescents and adults with a sport-related concussion (SRC).

Design Systematic review including risk of bias (modified Scottish Intercollegiate Guidelines Network tool).

Data sources MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Embase, APA PsycINFO, Cochrane Central Register of Controlled Trials, CINAHL Plus with Full Text, SPORTDiscus and Scopus searched until March 2022.

Study eligibility criteria (1) Original research including randomised controlled trials (RCTs), quasi-experimental designs, cohort, comparative effectiveness studies; (2) focus on SRC; (3) English; (4) peer-reviewed and (5) evaluated treatment.

Results 6533 studies were screened, 154 full texts reviewed and 13 met inclusion (10 RCTs, 1 quasi-experimental and 2 cohort studies; 1 high-quality study, 7 acceptable and 5 at high risk of bias). Interventions, comparisons, timing and outcomes varied, precluding meta-analysis. For adolescents and adults with dizziness, neck pain and/or headaches >10 days following concussion, individualised cervicovestibular rehabilitation may decrease time to return to sport compared with rest followed by gradual exertion (HR 3.91 (95% CI 1.34 to 11.34)) and when compared with a subtherapeutic intervention (HR 2.91 (95% CI 1.01 to 8.43)). For adolescents with vestibular symptoms/ impairments, vestibular rehabilitation may decrease time to medical clearance (vestibular rehab group 50.2 days (95% CI 39.9 to 60.4) compared with control 58.4 (95% CI 41.7 to 75.3) days). For adolescents with persisting symptoms >30 days, active rehabilitation and collaborative care may decrease symptoms.

Conclusions Cervicovestibular rehabilitation is recommended for adolescents and adults with dizziness, neck pain and/or headaches for >10 days. Vestibular rehabilitation (for adolescents with dizziness/vestibular impairments >5 days) and active rehabilitation and/ or collaborative care (for adolescents with persisting symptoms >30 days) may be of benefit.

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ The majority of individuals will recover during the initial 2–4 weeks following sport-related concussion (SRC), but up to 30% of adolescents and adults may have persisting symptoms and require rehabilitation.
- ⇒ A brief period of 24–48 hours of rest followed by return to activities of daily living and a return to sport and school strategy is recommended.
- ⇒ Subsymptom threshold aerobic exercise, cervicovestibular rehabilitation and collaborative care may be of benefit following SRC.

WHAT THIS STUDY ADDS

- ⇒ For adolescents and adults with symptoms of dizziness, neck pain and/or headaches for greater than 10 days, an individualised cervicovestibular rehabilitation programme is recommended.
- ⇒ For adolescents with dizziness for greater than 5 days, vestibular rehabilitation may be of benefit.
- ⇒ For adolescents with persisting symptoms of all types, active rehabilitation and collaborative care may be of benefit.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ Clinicians should recommend cervicovestibular rehabilitation for individuals who have sustained an SRC and report dizziness, neck pain and/or headaches for greater than 10 days.
- ⇒ Clinicians may also consider recommending vestibular rehabilitation (for adolescents with persisting dizziness for greater than 5 days), active rehabilitation and collaborative care for adolescents with persisting symptoms (>30 days) following SRC.

INTRODUCTION

Concussion is among the most commonly occurring injuries in sport.^{1–3} Sport-related concussion (SRC)



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results in a variety of symptoms and functional difficulties that limit the ability to undertake activities of daily living, school, work and sport and recreation.^{4–7} Fortunately, most concussion symptoms will resolve within the initial weeks following injury.⁸ However, up to 30% of children, adolescent and adults report persisting symptoms beyond 30 days.^{8–10}

As of the Berlin International Consensus on Concussion in Sport, consensus expert recommendations for the management of SRC include a brief period of relative cognitive and physical rest for 24–48 hours, followed by a gradual return to school/learn (RTSchool) and sport (RTSport).^{11–13} The RTSport strategy includes a general, gradual and progressive return to physical activities (PAs), with each step taking a minimum of 24 hours, provided that concussion-related symptoms do not increase.^{11–13} RTSchool and return to work should occur prior to final RTSport (eg, return to full competition) for children, adolescents and postsecondary students, although both can be progressed simultaneously. Medical clearance to RTSport is recommended prior to initiating sport with risk of contact/collision (typically step 5 of the RTSport strategy).¹¹ Once an individual has successfully completed the RTSchool/Learn and RTSport strategy, medical clearance to return to full sport can be considered, provided the athlete remains free of concussion-related symptoms at rest and exertion and no other clinical indications preclude RTSport.

In most cases, athletes of all ages progress well through the RTSport and, where relevant, RTSchool/Learn strategies, and have an uncomplicated recovery of activities. However, in up to 30% of cases of SRC, persisting symptoms may require further treatment/rehabilitation.^{8–12–14} Given the heterogeneous nature of SRC, patients experiencing different symptoms and functional limitations may respond to different treatments.^{12–15–16} Following the 5th International Conference on Concussion in Sport, recommendations for treatment/rehabilitation for individuals with ongoing symptoms following SRC included individually targeted cervical and vestibular rehabilitation and multifaceted collaborative care (eg, cognitive-behavioural therapy, pharmacotherapy and school accommodations).^{11–12–16} Some evidence also supported subsymptom threshold and submaximal exercise, with these interventions demonstrated to be safe when clinically supervised and directed.^{11–12} However, research evaluating treatment following concussion was recognised to be limited, especially for children.

Thus, the purpose of this systematic review was to evaluate the effects of targeted interventions following SRC. Rest and exercise interventions are described in another systematic review (Leddy *et al*¹⁵) and thus are not included in this review. We evaluated the effect of targeted interventions for cervical, vestibular, visual, headache, hormonal, cognitive, mood, behavioural and other problems, in facilitating recovery in children, adolescents and adults who have sustained a SRC.

METHODS

This systematic review follows a prespecified study protocol (PROSPERO 2022 CRD42022158814) and abides by the reporting standards recommended in the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹⁷ The authors amended their original study protocol by adding an assessment of the certainty/confidence in the overall body of evidence for each outcome assessment (ie, Grading of Recommendations Assessment, Development and Evaluation (GRADE) Working Group).¹⁸

Search strategy

The search focused on three main concepts: concussion, sport and rehabilitation/treatment. The search incorporated keywords, searched in specific fields (title, abstract, author supplied keywords) and subject headings. The author group provided keywords for treatment/rehabilitation, inclusion and exclusion criteria, and suggested database selection. A standardised search strategy for the terms “sport” and “concussion” was used across all systematic reviews, as defined in the methodology paper that accompanies this review (Schneider *et al*).¹⁹ An expert health sciences librarian (KAH) then developed the Medline search strategy incorporating the standard searches and the feedback from the author group for the third concept. The draft Medline search was pilot tested with key studies, as well as studies included in a similar review from fifth Consensus Statement on Concussion.¹²

The following databases were searched: MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations and Daily (Ovid), Embase (Ovid), APA PsycINFO (Ovid), Cochrane Central Register of Controlled Trials (Ovid), CINAHL Plus with Full Text (EBSCO), SPORTDiscus with Full Text (EBSCO) and Scopus (Elsevier). Searches were originally conducted October 2019. Because of the global pandemic, the originally proposed time for review completion was delayed 2 years, from 2020 to 2022. Thus, to ensure that the reviews were current for the consensus meeting, the search was rerun in March 2022. The search strategies for all databases are available as online supplemental file 1. The Medline search is annotated.²⁰

The draft Medline search was reviewed by the lead author and coauthors, and then peer-reviewed by an expert librarian (ZP). The revised Medline search was then translated for each database, with keywords similar across all databases, and subject headings responsive to the controlled vocabulary of the database (see online supplemental file 1—search strategy). Results from each database were exported as Research Information Systems (RIS) files and subsequently uploaded to Covidence for deduplication and screening.

Study eligibility

Studies needed to meet the following criteria to be eligible: (1) be original research (including randomised controlled trials (RCTs), quasi-experimental designs, cohort, comparative effectiveness studies); (2) have SRC as the focus of the study (ie, >50% of the sample sustained a concussion from sport participation); (3) be written in English; (4) be peer-reviewed and (5) evaluate the effect of treatment of cervical, vestibular, auricular, visual, headache, hormonal, cognitive, mood, behavioural, or other problems, or other interventions following SRC. Studies were excluded if they (1) were review articles, commentaries, editorials; (2) were published in abstract form only; (3) were books, book chapters, conference abstracts or dissertations/theses and (4) studies older than 2001.

Study screening

A rapid screen to remove any records that were irrelevant to the topic (ie, non-human studies, non-concussion, conference proceedings, opinion papers) was completed by two authors prior to study selection. Authors independently screened 50 citations while blinded to another author designated as a methods author to assess inter-rater agreement for the title and abstract screen. Clarification of the inclusion/exclusion criteria occurred if the inter-rater agreement was less than 80%. The title and abstract screen was then performed independently by two

authors per citation, one being the methods author and the other a coauthor. All coauthors were invited to identify any additional citations for review at the full text stage. Full texts were obtained by a research assistant and full text screening was completed independently in Covidence by two authors, including reasons for exclusion.

Data extraction

A data extraction (DE) table was generated a priori with contributions and approval from all authors. The a priori DE table was piloted and further reviewed by the authorship team. Each citation included in the full text phase underwent data extraction and risk of bias (ROB) assessment by two authors independently—one author designated as a methods author

(MC, KJS) and one as coauthor (VA, GAD, CD, IG, CLM, JP, JKR-M, RSM and KOY). An adapted Scottish Intercollegiate Guidelines Network tool²¹ for the content area of SRC for each design type was used as the ROB tool for each review (Schneider *et al*).¹⁹ All authors underwent a standardised DE and ROB training process (both overall and specific to this review) prior to initiating DE and ROB assessment. Any discrepancies were discussed between reviewing authors until consensus was reached. Where needed, the lead author acted to resolve conflicts. Any author of the systematic review who was an author on one of the included studies was not involved in the data extraction, ROB or resolving conflicts for their own work. A rating of the overall quality of evidence was assigned for each subtopic area identified using GRADE.¹⁸

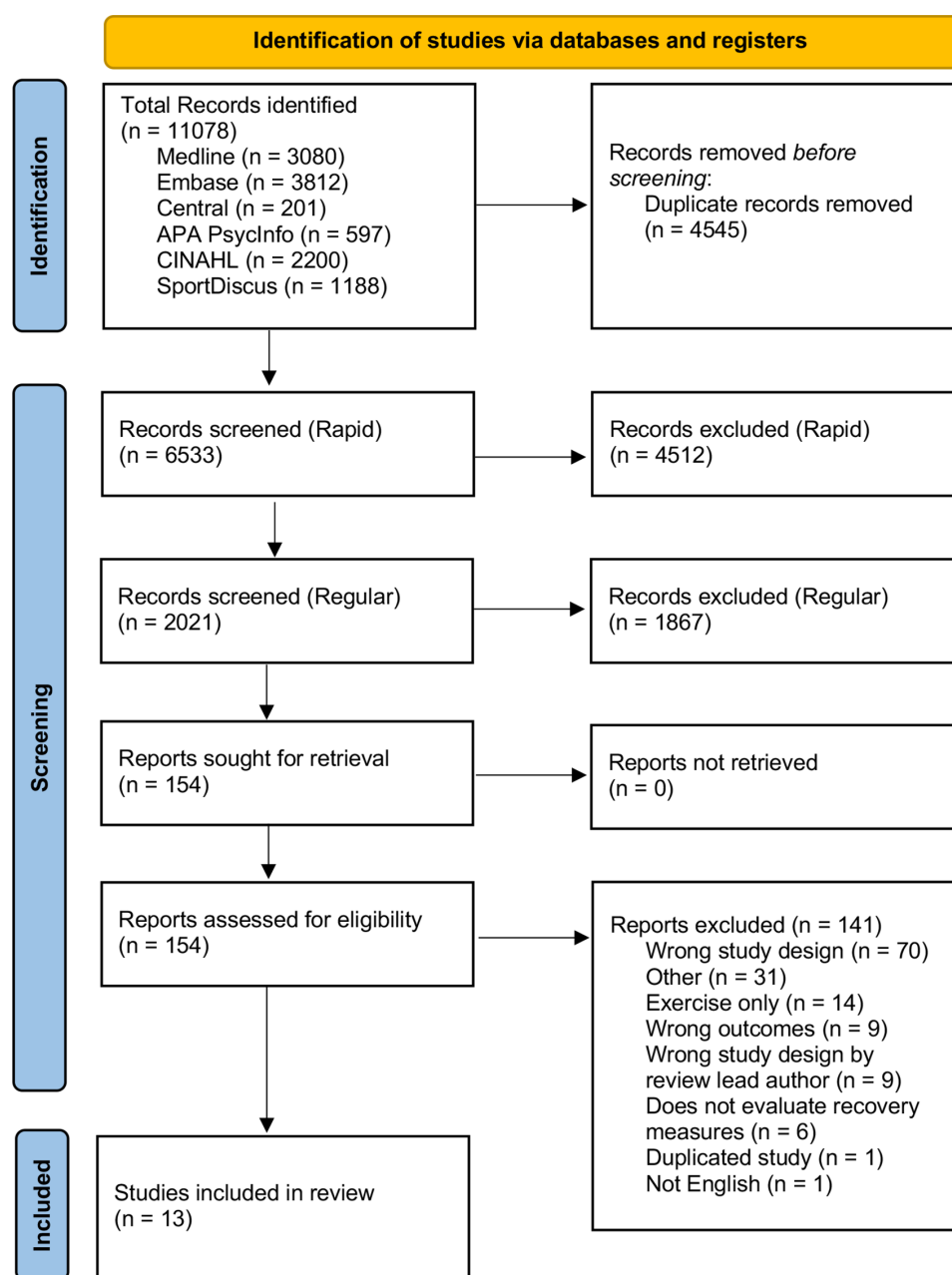


Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 flow diagram for new systematic reviews which included searches of databases and registers only. *Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers). From: Page *et al*.¹⁷ APA, American Psychological Association; CINAHL, Cumulated Index to Nursing and Allied Health Literature; n, number.

Table 1 Data extraction for included studies

Author, year	Study design	Participants (n)	Primary outcome of recovery	Timing of intervention	Intervention description	Control description	Risk of Bias (++, +, -)
Active rehabilitation							
Gauvin-Lepage, 2018 ²³	Cohort	n=355 (188 girls, 167 boys) aged 8–17 years	Severity of PCSS (/132)	3–4 weeks postinjury if still symptomatic	AR protocol and other interventions as needed	N/A	(+)
Chan, 2017 ²²	RCT	n=19, 14 female, 5 male; mean age=15.5; SD 1.47.	PCSS	≥4 weeks postinjury; followed for 6 weeks	AR protocol (Gagnon <i>et al</i> ⁵² 2009)+TAU	TAU	(+)
Cervical/vestibular rehabilitation							
Reneker, 2017 ²⁶	RCT	IG n=22, mean age 16.5 (SD 2.9) years, 14 males, 8 females; CG n=19, mean age 15.9 (SD 2.9) years, 11 males, 8 females	(1) Symptom recovery: (2) Medical clearance to return to light aerobic exercising	Initiated starting at 10 days postconcussion	Cervical and vestibular physiotherapy	Subtherapeutic treatment with same physiotherapists	(+)
Kontos, 2021 ²⁷	RCT	VESTIB (n=25, mean age 15.3±1.6, 16 female; CONTROL n=25, mean age 15.3±1.7, 15 female)	VOMS	5.7±4.0 days postinjury VESTIB and 6.6±4.0 days postinjury CONTROL	4 weeks vestibular intervention group	4 weeks behavioural management control group	(+)
Schneider, 2014 ¹⁴	RCT	n=31 included participants; (12–30 years; 18 male and 13 female)	Medical clearance to return to sport	10 days or greater postinjury	Individualised cervical and vestibular rehabilitation	ROM exercises, stretching and postural education followed by graded exertion	(++)
Collaborative care							
McCarty, 2016 ¹⁶	RCT	n=49; mean age 15.0 years (range 11–17); 65% female (n=32), 35% male (n=17)	HBI, PHQ-9, PROMIS-PA8, PedsQL	Median (IQR) 60 (38) collaborative care group and 74.5 (59.5) days following concussion usual care group.	Collaborative care that included care coordination and management	Care as usual	(+)
McCarty, 2021 ²⁸	RCT	n=200, Mean age 14.7 (1.7) years; 124 girls (62.0%); n=99 usual care; n=101 collaborative care.	HBI, PedsQL, Revised Child Anxiety and Depression Scale-Short Version and GAD-7 and PHQ-9	At least 1-month postinjury and within 9 months of injury	Collaborative care including cognitive behavioural therapy and care management	Care as usual	(+)
Supplements							
Walter, 2017 ²⁹	RCT	n=42, Collegiate athletes (18–24 years; 20 male and 22 female)	PCSS HeadRehab VR system: HVLT-R, EEG	6 months to 3 years postconcussion	Enzogenol capsules (ENZO Nutraceutical), 1000mg daily	Non-differentiable placebo, 1000 mg daily	(+)

+, acceptable ROB; ++, high quality/low ROB; -, low quality/high ROB; AR, active rehabilitation; EEG, electroencephalogram; GAD-7, generalised anxiety disorder assessment; HBI, health and behaviour inventory; IG, intervention group; PCSS, post concussion symptom scale; PedsQL, paediatric quality of life; PHQ-9, patient health questionnaire; PROMIS-PA8, patient-reported outcomes measurement information system; RCTs, randomised controlled trials; TAU, treatment as usual; VOMS, vestibular ocular motor screen.

Analysis/synthesis

Recognising the heterogeneity in patient populations, interventions evaluated and study designs, a formal meta-analysis was not considered. Rather a narrative synthesis by treatment type was conducted and summarised graphically.

Equity, diversity and inclusion statement

All eligible studies were included in this review regardless of sample characteristics. Of note, most studies included both males and females and no studies specifically mentioned the inclusion of para athletes. The authors of this review included 10 women and 9 men across a range of disciplines (ie, physiotherapists, physicians, neuropsychologists, scientists, epidemiologists, athletic trainers and librarians) and career stages (graduate students, early-career, mid-career and senior career researchers and clinicians).

RESULTS

A total of 6533 studies were screened for inclusion in this systematic review (see online supplemental file 1 and figure 1). One hundred and fifty-four studies were reviewed in full text. Thirteen studies were included in the review (online supplemental file 3) and evaluated the following types of treatment: active rehabilitation (AR)^{22 23} (defined as including submaximal

aerobic training, light coordination and sport-specific exercises, visualisation and imagery techniques and a home exercise programme), cognitive activity,²⁴ medication,²⁵ cervical and/or vestibular rehabilitation,^{14 26 27} collaborative care,^{16 28} supplements,^{29 30} head and neck cooling³¹ and osteopathic manipulative medicine (OMM).³² Three were completed in Canada and 10 in the USA. Ten RCTs, one quasi-experimental study and two cohort studies were included in the review. The studies varied in quality with one RCT rated as high quality/low ROB, eight studies as acceptable and four as low quality/high ROB. Studies that were acceptable or high quality/low ROB are summarised in table 1. Nine studies included children (up to age 12), 11 included adolescents (age 13–18 years) and 6 included adults (>18 years). One study did not specify sex of the participants and the remaining studies included both females and males. A total of 634 females and 624 males participated in the included studies that reported sex. No studies specifically described inclusion of para athletes. Two studies evaluated treatment in collegiate sport athletes^{29 32} and the remaining studies did not specify a specific level of sport. Eleven studies recruited from specialty clinics (sport medicine, concussion, rehabilitation), one from the emergency department³⁰ and one did not specify.²⁹ Four studies recruited subjects within 1 week following concussion, five between 1 week and 4 weeks, and four greater than 1 month

Collaborative Care	McCarty et al 2016	11–17 years
	McCarty et al 2021	11–18 years
Vestibular Rehabilitation	Kontos et al 2021	12–18 years
Cervicovestibular Rehabilitation	Reneker et al 2017	10–23 years
	Schneider et al 2014	12–30 years
Active Rehab	Gauvin-Lepage et al 2018	8–17 years
	Chan et al 2017	12–18 years
Enzogenol	Walters et al 2017	18–24 years

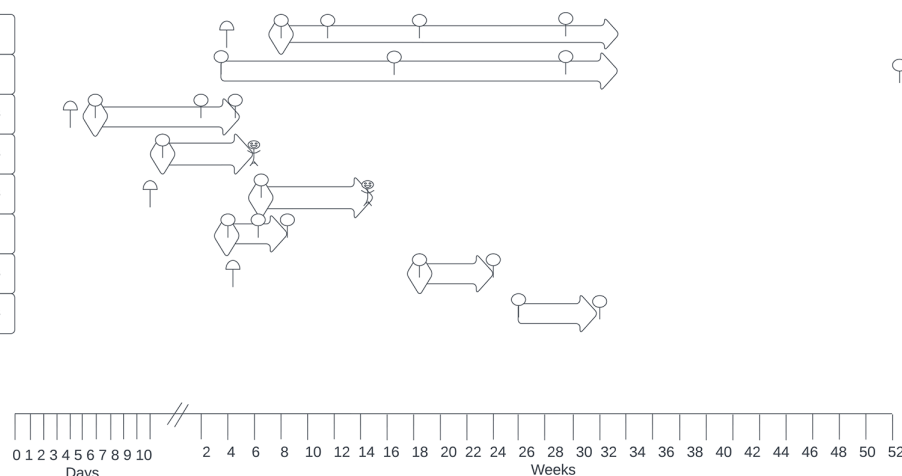


Figure 2 † Earliest time point for treatment initiation. ♦ Mean or median time to initiation of treatment; where no mean or median is available, intervention period starts at the earliest time point. ▬ Intervention period. ♣ Assessment time point. ✕ Intervention endpoint at time of medical clearance or end of trial for reassessment.

postinjury. Most studies ($n=10$) included participants based on diagnosis of SRC (using variable definitions and three studies lacking a definition) and the presence of symptoms or persisting symptoms. Two RCTs evaluating cervicovestibular (CV) rehabilitation included participants based on the presence of specific symptoms of dizziness²⁶ or dizziness/neck pain/headaches¹⁴ or objective findings/impairments of vestibular/balance and/or cervical spine function.^{14 26} One RCT evaluated vestibular rehabilitation in adolescents/adults with vestibular symptoms and objective findings.²⁷ A narrative synthesis by treatment type is included below and summarised graphically in figure 2 with the full data extraction table available online (supplemental file 2).

Active rehabilitation

Two included studies evaluated the effect of AR in children and adolescents. Girls reported higher symptoms across all three time points (28.5, 20.8 and 15.3 at 0, 2 weeks, 4 weeks, respectively, for girls and 19.96, 11.5 and 10.90 for boys), but the rate of recovery over time was similar in boys and girls.³³ In an RCT, adolescents aged 12–18 years with persisting symptoms for greater than 1 month following SRC who participated in AR had a greater decrease in Post Concussion Symptom Score (PCSS) scores compared with those who participated in treatment as usual (symptom management, advice on return to play, facilitation of RTSchool and a psychiatry consultation) ((Wald's $t=2.15$, $p=0.047$).²²

Vestibular rehabilitation

One RCT compared precision vestibular rehabilitation (individualised vestibular rehabilitation exercises for 30 min daily) in addition to behavioural management (which included PA, sleep, hydration, nutritional and stress management) to behavioural management plus stretching and PAs (control group) in athletes aged 12–18 years with vestibular symptoms or impairments.²⁷ The primary outcome of Vestibular Ocular Motor Screen subcomponent scores improved for horizontal (1.63; 95% CI -3.20 to 0.06 ; $p=0.04$) and vertical (2.24; 95% CI -4.01 to 0.48 ; $p=0.01$) vestibulo-ocular reflex (VOR) mean differences. In this study, the mean days to medical clearance for intervention group was 50.2 (95% CI 39.9 to 60.4) days vs 58.4 (95% CI 41.7 to 75.3) days in the control group.

Cervical and vestibular rehabilitation

Two RCTs evaluated the effects of CV rehabilitation/physiotherapy compared with subtherapeutic/general exercise prescription. Cervical and vestibular physical therapy for dizziness in children, adolescents and adults (aged 10–23 years) with dizziness of at least 3 on a scale of 0–6 was compared with a sham, subtherapeutic or non-progressive exercises intervention starting at 10 days postconcussion.²⁶ The experimental group achieved medical clearance to return to light aerobic exercise faster than the control group (HR 2.91; 95% CI 1.01 to 8.43).²⁶ Another RCT evaluated the effects of individualised CV rehabilitation compared with rest followed by gradual exertion in adolescents and adults with dizziness, neck pain and/or headaches, and found that participants in the CV group were 3.91 (95% CI 1.34 to 11.34) times more likely to be medically cleared within the 8-week intervention period than control participants.¹⁴

Collaborative care

Two RCTs evaluated the effect of collaborative care (including care coordination/management, cognitive-behavioural therapy and multidisciplinary case conferences, with enhanced medication consultation if warranted) compared with usual care (an initial visit at the clinic, and care as they would typically receive at the clinic including imaging, referral to rehabilitation medicine, neuropsychological testing, prescribed sleep medication as needed and potentially also subthreshold exercise prescription from a physical therapist) in adolescents with persisting symptoms for more than 1 month following concussion.^{16 28} A pilot RCT in children and adolescents aged 11–17 (57.5% with SRC) found improvement in postconcussive symptoms at 6 months (13% of intervention patients vs 41.7% of control participants symptomatic at 6 months). The intervention group was 'very satisfied' with care 82.6% of the time compared with 45.8% of time in the control group ($p=0.001$).¹⁶ Of note, 18.8% of the sample reported a diagnosis of attention-deficit hyperactivity disorder (ADHD) or learning disability (LD) and 37.5% reported a history of anxiety or depression. A second RCT (88% with SRC) by a similar group of investigators found that collaborative care resulted in improved Health and Behaviour Inventory (HBI) scores at 3 months (3.4 point decrease; 95% CI 0.1 to 6.6 point decrease) and 12 months (4.1 point decrease; 95% CI

0.4 to point decrease) compared with usual care.²⁸ In addition, Paediatric Quality of Life scores at 12 months improved by a mean of 4.7 points (95%CI 0.05 to 9.3 points) in the intervention group compared with the control group. Sleep quality improved at 3, 6 and 12 months in the intervention group (HBI change scores of 0.3 (0.2–0.5), 0.2 (0.04–0.5) and 0.4 (0.2–0.6), respectively. There was no difference in other outcomes.²⁸ Preinjury anxiety (31.0%), depression (18.0%) and chronic headaches (25.0%) were prevalent in this sample.

Supplements

Two RCTs evaluated supplements as a potential treatment for SRC. In an acceptable quality study, Enzogenol supplementation (1000 mg daily) when compared with a placebo in collegiate athletes (aged 18–24 years) who had ongoing symptoms for 6 months to 3 years following concussion did not alter time to recovery.²⁹ In a study at high ROB, magnesium (PO 400 mg two times per day) and acetaminophen (or paracetamol) (500 mg two times per day) as a treatment for acute concussion (within 48 hours of injury) in adolescents 12–18 years was compared with acetaminophen (or paracetamol) alone. No differences between groups were identified.³⁰

Other

One quasi-experimental study evaluated the effect of amantadine and found a greater decrease in symptoms the amantadine group than the controls ($F=8.72$, $p=0.005$); however, the amantadine group had greater symptoms and scored worse on verbal and visual memory and reported more symptoms at the start of the study which was not accounted for in the analysis.²⁵ The effect of cognitive activity was evaluated in one prospective cohort study,²⁴ which reported that higher initial total PCSS score (OR 0.98; 95%CI 0.97 to 0.99) and engaging in a relatively high degree of cognitive activity (highest quartile for 'cognitive activity days'; OR 0.99 (95% CI 0.99 to 0.996) were associated with a longer duration of symptoms compared with the other three quartiles.²⁴ An RCT comparing the use of a head and neck cooling device in adolescents aged 12–17 years at the initial intake (0–8 days postinjury) and again 72 hours later compared with standard of care treatment found a faster recovery in symptom severity score in athletes who received the treatment ($p=0.003$).³¹ An RCT with high ROB compared OMM to concussion education in adults with ongoing symptoms who were within 2 weeks of concussion and found a greater decrease in the number of symptoms (-3.93 ± 3.92 vs -0.64 ± 0.97 ; $p=0.002$) and severity of symptoms in the intervention group (-17.33 ± 17.38) than the concussion education group (-3.64 ± 5.89 ; $p=0.001$).³²

Quality of evidence (GRADE)

Most subcategories of treatment types had just one study and thus were unable to demonstrate consistency, had small effect sizes and had acceptable or high ROB, resulting in a GRADE quality rating of very low (1/4) (see online supplemental file 4). CV rehabilitation was rated as low based on two RCTs (one with low ROB and one with acceptable quality) with large effect sizes but wide, overlapping CIs; because there were only two studies, the GRADE was downgraded two points to low.

DISCUSSION

This systematic review included 13 studies evaluating treatments following concussion, 9 of which were of either high quality or of acceptable quality. The studies were heterogeneous in terms of types of treatments evaluated, timing of intervention, sample

recruited, age of participants and outcome measure selected. This heterogeneity precluded meta-analyses. Of the high quality and acceptable studies, AR, CV rehabilitation, vestibular rehabilitation, and collaborative care all demonstrated positive effects on recovery. To date, we cannot determine whether interventions tailored to a specific symptom (eg, most prominent symptom) are more effective than a one-size-fits-all treatment approach. Most included studies did not limit recruitment to patients with a particular symptom profile or clinical feature. Subgroups of patients within the included samples may have been more likely to respond to treatment than others. Thus, the effect of intervention may reflect the individual patient characteristics and the included samples, such as the included study evaluating vestibular rehabilitation alone (Kontos *et al*) only including patients with vestibular symptoms/impairments. Notably, in adolescents with persisting symptoms, the two studies evaluating collaborative care included samples with high prevalence of ADHD, LD, chronic headaches and a history of anxiety or depression.²⁵

An AR intervention including submaximal aerobic training, light coordination and sport-specific exercises, visualisation and imagery techniques, and a home exercise programme in children and adolescents presenting to a concussion clinic at a paediatric rehabilitation hospital found greater reductions in overall symptom burden in the AR group compared with the control group ($n=19$)²² and similar rates of recovery in girls and boys.³³ The effect of early introduction (within days) of light PA (such as walking) as well as prescribed subsymptom threshold aerobic exercise treatment on concussion recovery is the focus of a separate systematic review and supports the inclusion of PA/prescribed exercise as a safe and effective rehabilitation approach following SRC (Leddy *et al*).³⁴

In three studies, the presence of specific symptoms (dizziness for all three studies and dizziness, neck pain and/or headaches for one) and clinical findings was an inclusion criterion. These studies demonstrated significant improvements with cervical and vestibular rehabilitation^{26 35} or with vestibular rehabilitation alone.²⁷ These interventions included individualised treatments based on patient presentation. Altered vestibulo-ocular, balance function and cervical spine function have been reported following concussion³⁶ and, in some cases, the presence of these findings has negatively affected prognosis (ie, predicted longer recovery from concussion).^{37 38} There is a large body of literature demonstrating safety and efficacy of vestibular rehabilitation for individuals with different types of vestibular disorders.^{39 40} The same can be said for headaches and cervical spine pain.^{41–44} While the studies included in the current review demonstrated a positive impact on symptoms and recovery in adolescents and adults with SRC, future research to understand the effect of cervical and vestibular rehabilitation in the presence of different types of vestibular, cervical spine and headache disorders is warranted. Of interest, vestibular symptoms and impairments and cervical spine findings and headaches have been reported concomitantly 46%–93% of the time.^{14 35 45} For example, in a prospective cohort study in mild Traumatic Brain Injury (mTBI) of all types, symptom provocation was common during oculomotor and VOR tests despite normal test outcomes, with high proportions of this group reporting cervical spine pain or presenting with an abnormal cervical spine evaluation.⁴⁶ The presence of cervical spine findings does not exclude the presence of concussion, but rather cervical spine injury could occur concomitantly with concussion.⁴⁷ Future research to understand the underlying pathophysiology of cervical spine pain, sensorimotor and neuromuscular control changes observed following concussion is needed.

Given the heterogeneity of symptoms and impairments following SRC, interventions designed to address overall symptom burden such as AR and collaborative care may benefit most patients with SRC. For those who have specific symptoms of dizziness±neck pain and/or headaches, targeted vestibular rehabilitation or CV rehabilitation may reduce symptoms and time to medical clearance to RTSport. Other interventions such as head and neck cooling, supplements and other novel treatments require further evaluation and may be of benefit, but the existing studies included in this review had high ROB. An approach of matching interventions to patient presentation in the presence of persisting symptoms may lead to improved outcomes but requires further evaluation.⁴⁸ The timing of treatment, combinations of treatment, and effect of previous concussion(s) on outcomes is not well understood and requires evaluation.

Limitations

Since our previous review,¹² studies with higher quality designs have appeared, but there remain many limitations and areas to address in future research. Many of the included studies did not describe the treatment components and how they were delivered in sufficient detail. Consistent use of reporting guidelines such as the Template for Intervention Description and Replication⁴⁹ would be helpful. Few studies described the theoretical basis for their treatment design or precisely specified the treatment targets and proposed mechanisms of action. Treatment development frameworks such as the Rehabilitation Treatment Specification System⁵⁰ should be considered. Most studies evaluated multifaceted treatments, leaving unanswered questions about the contribution of specific treatment components to the overall intervention benefit. Decomposition studies could help isolate 'active ingredients'. In addition, minimal evidence exists on the implementation and translation of these interventions into real world practice, highlighting the need for future research on effectiveness.

An understanding of effects of age and sex on treatment was the focus of just one study.³³ In addition, most studies do not include children below the age of 12, para athletes, or consider gender, ethnicity, socioeconomic status or other social determinants that may influence outcome. Other factors may modify treatment outcome such as a history of concussion, presence of anxiety, sleep disturbance and presence of migraine headaches. Most samples were recruited from specialty clinics and one from the emergency department, resulting in selection bias.

The optimal timing of intervention is unclear. The included studies focus on various time points following injury, making it difficult to compare across the spectrum of recovery. The same interventions have not been compared at more than one time point. Notably, the second phase of one of the RCTs included in this systematic review evaluated outcomes in study participants who had not recovered and crossed over into the treatment they had not been initially randomised to (either CV rehabilitation or control arm).³⁵ While only the most severely symptomatic participants engaged in this phase of the trial, 33% who crossed into the treatment group and 0% in the control group were medically cleared to RTSport within 8 weeks of initiating treatment—both of which were lower proportions than in the original phase of the study.³⁵ It may be that earlier treatment is more beneficial. However, future research is needed to identify the right treatment for the right patient at the right time.

Some treatment studies relied heavily on self-reported symptoms as a primary outcome. Self-reported outcomes are prone to bias, especially in studies evaluating behavioural interventions,

where complete participant blinding is not possible. The field would benefit from validation of objective endpoints, especially endpoints mechanistically linked to the interventions under study. Multidimensional endpoints that include (1) patient-reported outcomes, (2) objective measures and (3) an operationalised definition of medical clearance to RTSport would provide a more comprehensive assessment of treatment benefit. Ideally, this would be done using common data elements to facilitate comparisons across studies and enabling future meta-analyses.⁵¹

Our review included 10 RCTs. However, in some cases, methods for allocation concealment or randomisation were not well described, leading to the potential of a selection bias. Misclassification bias was also a threat due to the subjective nature of outcomes reported and cointervention that may have affected outcomes. Future study should also record PA, cognitive activity and cointerventions, which may also influence recovery.

CONCLUSIONS

The evidence supports CV rehabilitation in adolescents and adults with symptoms of dizziness±headaches and/or cervical spine pain in the subacute time period (10 days and greater) to facilitate resolution of symptoms and improved function following SRC. Vestibular rehabilitation (for vestibular symptoms/impairments >5 days postinjury), AR and collaborative care (when symptoms persist for >30 days) may be of benefit following concussion for children and adolescents. Future research to better understand subcomponents of treatment as active ingredients and the optimal timing and type of rehabilitation—individually and in combination of rehabilitation is needed.

Key recommendations

- ⇒ For adolescents and adults with dizziness, neck pain and/or headaches for greater than 10 days following concussion, rehabilitation should include an individualised approach including cervicovestibular rehabilitation which may decrease the time to medical clearance to return to light aerobic activity (HR 2.91 (95% CI 1.01 to 8.43)) and medical clearance to return to sport (HR 3.91 (95% CI 1.34 to 11.34)).
- ⇒ For adolescents with persisting vestibular symptoms or impairments, vestibular rehabilitation may be of benefit to decrease symptoms and time to medical clearance to return to sport.
- ⇒ For adolescents with persisting symptoms for greater than 30 days, active rehabilitation and/or collaborative care may decrease symptoms.
- ⇒ Future study should consider (1) the inclusion of patient-reported outcomes, specific objective outcomes and operationally defined functional measures of recovery; (2) general versus specific/targeted treatments; (3) recording of specific treatment subcomponents; (4) timing; (5) frequency; (6) duration; (7) intensity; (8) combinations of treatments; (9) age and (10) sex, gender, ethnicity, socioeconomic status, para athlete-specific considerations.

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